The Islamia University of Bahawalpur
Department of Information and Communication Engineering
BS Cyber Security and Digital Forensics

Cloud Security (Lab – 03)
Implementing Enterprise Governance with Azure AD using MFA, Conditional Access, and Role-Based Access Control

Objective

This lab will introduce enterprise governance strategies in Azure, focusing on:

- Role-Based Access Control (RBAC)
- Configuring Multi-Factor Authentication (MFA)
- Setting up Conditional Access Policies
- Implementing Azure AD Identity Protection

Lab Overview

Azure Active Directory (Azure AD) is Microsoft's cloud-based identity and access management service, crucial for managing access to Azure resources and applications. This lab will cover creating users, assigning roles, configuring MFA, and setting Conditional Access Policies to enhance security.

Tools Required

- Microsoft Azure Account: Access to the Azure portal.
- Azure Active Directory: To manage users and enforce identity protection.
- Azure AD Premium P1 or P2 license (for full Conditional Access and Identity Protection capabilities).

Steps to Complete the Lab

Add a New User

You can create a new user using the Azure Active Directory portal. To add a new user, follow these steps:

1. Sign in to the Azure portal in the User Administrator role for the organization.
2. Search for and select Azure Active Directory from any page.
3. Select Users, and then select New user.

On the User page, enter information for this user:

- Name (Required): The first and last name of the new user. For example, Mary Parker.
- User Name (Required): The username of the new user. The domain part of the user name must use either the initial default domain name, <yourdomainname>.onmicrosoft.com, or a custom domain name, such as contoso.com.

Groups. Optionally, you can add the user to one or more existing groups. You can also add the user to groups later.
Directory role: If you require Azure AD administrative permissions for the user, you can add them to an Azure AD role. Y
Job info: You can add more information about the user here or do it later.

Add a New Guest User

You can also invite the new guest user to collaborate with your organization by selecting Invite user from the New user page. If your organization's external collaboration settings are configured such that you're allowed to invite guests, the user will be emailed an invitation they must accept in order to begin collaborating.

Add a New User within a Hybrid Environment

If you have an environment with both Azure Active Directory (cloud) and Windows Server Active Directory (on-premises), you can add new users by syncing the existing user account data.

Delete a User

To delete a user, follow these steps:

1. Sign in to the Azure portal using a User administrator account for the organization.
2. Search for and select Azure Active Directory from any page.
3. Search for and select the user you want to delete from your Azure AD tenant. For example, Mary Parker.
4. Select Delete user.

The user is deleted and no longer appears on the Users - All users' page. The user can be seen on the Deleted users page for the next 30 days and can be restored during that time.

You can delete an existing user using Azure Active Directory portal.

Assign administrator and non-administrator roles to users with Azure Active Directory

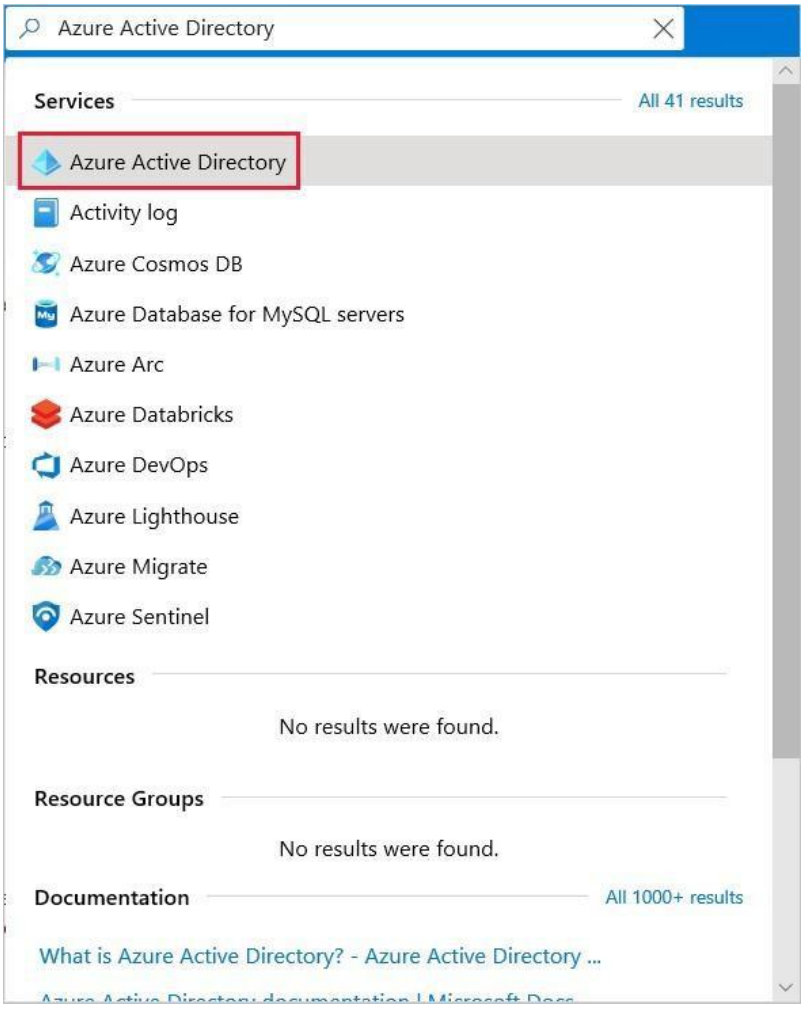
In Azure Active Directory (Azure AD), if one of your users' needs permission to manage Azure AD resources, you must assign them to a role that provides the permissions they need.

Assign roles

A common way to assign Azure AD roles to a user is on the Assigned roles page for a user.

1. Go to the Azure portal and sign in using a Global administrator account for the directory.

2. Search for and select Azure Active Directory.



3. Select Users.
4. Search for and select the user getting the role assignment. For example, Demo.
5. On the Demo - Profile page, select Assigned roles. The Demo – Administrative roles page appears.
6. Select Add assignments, select the role to assign to Alain (for example, Application administrator), and then choose Select.

The Application administrator role is assigned to Alain Charon, and it appears on the Alain Charon - Administrative roles page.

Remove a Role Assignment

If you need to remove the role assignment from a user, you can also do that from the Alain Charon - Administrative roles page.

1. Select Azure Active Directory, select Users, and then search for and select the user getting the role assignment removed. For example, Alain Charon.
2. Select Assigned roles, select Application administrator, and then select Remove assignment.

Configuring Multi-Factor Authentication (MFA)

1. Enable MFA for a User

- Navigate to Azure Active Directory > Users.
- Select Multi-Factor Authentication.
- In the MFA portal, select the user you want to enable MFA for, and click Enable.
- Instruct the user to complete MFA setup on their next login.

2. Set Up Additional Security Settings
   - In Azure Active Directory, navigate to Security > Multi-Factor Authentication > Additional cloud-based MFA settings.
   - Customize MFA options, including phone call, SMS, and app-based authentication.

Implementing Conditional Access Policies

1. Create a Conditional Access Policy
   - Go to Azure Active Directory > Security > Conditional Access > New policy.
   - Name your policy (e.g., "Location-Based Access").
2. Define Policy Conditions
   - Users or workload identities: Choose users to whom the policy will apply.
   - Cloud apps or actions: Select the applications the policy will apply to (e.g., Microsoft 365).
   - Conditions: Configure conditions based on Location or Device state.
3. Set Access Controls
   - Under Grant, select Require Multi-Factor Authentication and Require device to be marked as compliant.
   - Click On under Enable Policy and save the policy.
4. Testing and Validating the Conditional Access Policy
   - Log in from a location that should trigger the policy to validate if MFA is enforced correctly.

Managing Users and Groups in Azure AD

1. Add a New User to Azure AD
   - Navigate to Azure Active Directory > Users > New user.
   - Enter the user's details, including their Name, User name, and Directory role if needed.
   - Assign the user to any relevant Groups as needed, then click Create.
2. Delete a User
   - In Azure Active Directory > Users, search for and select the user to delete.
   - Select Delete user. Deleted users are moved to the Deleted users page for 30 days, where they can be restored.
3. Add a New Guest User
   - In Azure Active Directory > Users, select New guest user.
   - Enter the guest's Email address and select Invite to send an email invitation.

Review and Lockdown Resource Groups

1. Create a Resource Group
   - Navigate to Resource groups > Add.
   - Enter a Resource Group Name and select the Region.
   - Click Review + create and then Create.
2. Apply a Lock to the Resource Group

- Open the resource group created.
- In the left pane, select Locks > Add.
- Set a lock name, choose Read-only or Delete as the lock type, and add any relevant notes.
- Save the lock settings to prevent accidental deletion or modification of critical resources.

Verification and Testing

1. Test MFA and Conditional Access Policies
   - Attempt to access the Azure portal using a user account with Conditional Access policies applied to ensure MFA or access restrictions are triggered.
2. Verify Role Assignments and Access Controls
   - Log in as a user with assigned roles and check access permissions for various Azure resources to ensure appropriate restrictions.

Viva Voce

1. What are the main components of Azure AD, and how does it improve enterprise security?
2. How do role-based access controls help to secure resources in Azure?
3. Describe the steps to configure MFA in Azure. Why is MFA important?
4. Explain the process and benefits of Conditional Access policies.
5. What is the purpose of adding locks to a resource group?
6. What factors should be considered when assigning roles in Azure AD?